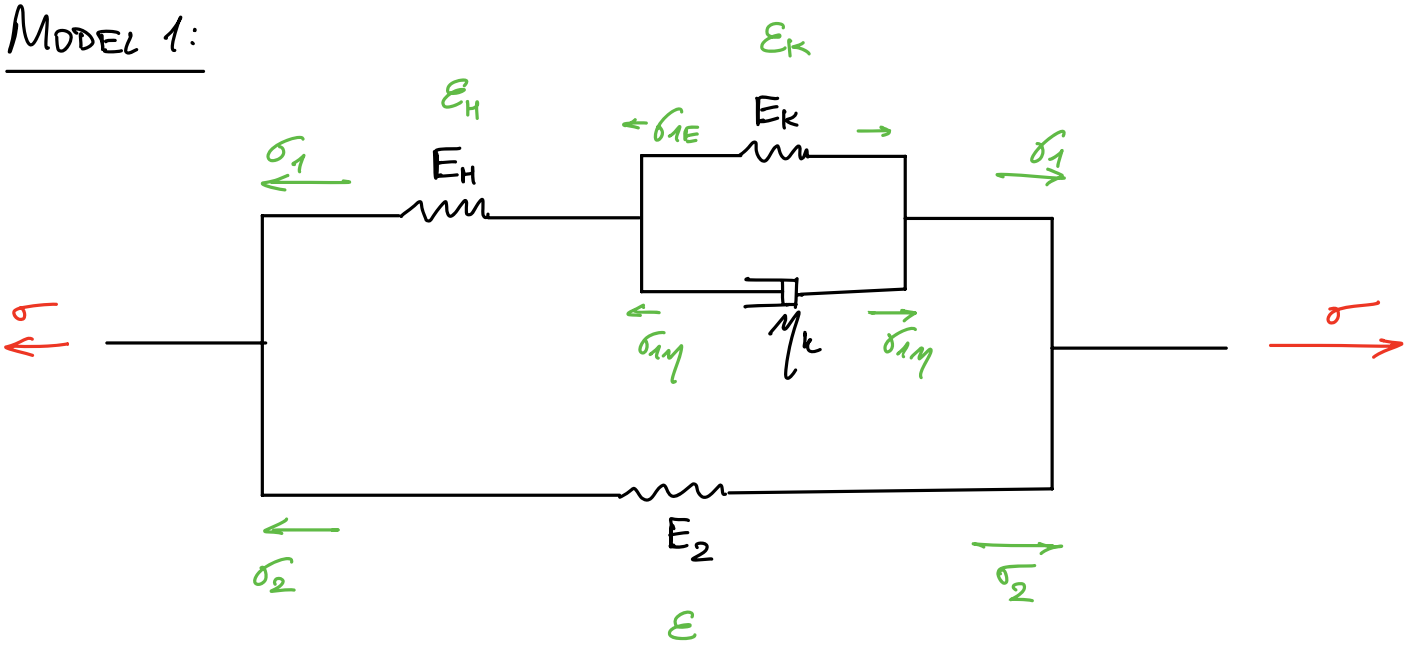
4 Parameter Rheological model :

Model 1:



* Stress in the branches : $\sigma = \sigma_1 + \sigma_2$ → (1)
* Strain in the branches : ε
* Branch 2 : $\sigma_2 = E_2 \varepsilon$ → (2)
* Branch 1 strains : $\varepsilon = \varepsilon_H + \varepsilon_K$ → (3)
* Branch 1 stresses :

$$\left.\begin{aligned} \sigma_1 &= \sigma_{1E} + \sigma_{1\eta} \\ \sigma_{1E} &= E_K \varepsilon_K \\ \sigma_{1\eta} &= \eta_k \dot{\varepsilon}_K \end{aligned}\right\} \rightarrow \sigma_1 = E_K \varepsilon_K + \eta_k \dot{\varepsilon}_K \rightarrow (4)$$

$$\sigma_1 = E_H \varepsilon_H \rightarrow (5)$$

* Solving to obtain everything in terms of $\varepsilon, \dot{\varepsilon}, \sigma, \dot{\sigma}$:

From (5) $\varepsilon_H = \dfrac{\sigma_1}{E_H}$ From (3) $\varepsilon_K = \varepsilon - \varepsilon_H = \varepsilon - \dfrac{\sigma_1}{E_H}$

Inserting into ④ $\quad \sigma_1 = E_K\left(\varepsilon - \frac{\sigma_1}{E_H}\right) + \eta_K\left(\dot{\varepsilon} - \frac{\dot{\sigma}_1}{E_H}\right)$

$$\dot{\sigma}_1(\eta_K) + \sigma_1(E_H + E_K) = \dot{\varepsilon}(E_H \eta_K) + \varepsilon(E_K E_H) \longrightarrow (6)$$

From ① & ② $\quad \sigma_1 = \sigma - \sigma_2 \Rightarrow \sigma_1 = \sigma - E_2\varepsilon$

$$(\dot{\sigma} - E_2\dot{\varepsilon})\eta_K + (\sigma - E_2\varepsilon)(E_H + E_K) = \dot{\varepsilon}(E_H \eta_K) + \varepsilon(E_K E_H)$$

* Collecting the terms we obtain

$$\dot{\sigma}\eta_K + \sigma(E_H + E_K) = \varepsilon\left(E_2(E_H + E_K) + E_K E_H\right) + \dot{\varepsilon}\left(\eta_K(E_H + E_2)\right) \longrightarrow (7)$$

* Creep behaviour :

$\sigma = \tilde{\sigma}$, $\dot{\sigma} = 0$

$$\therefore \quad \dot{\varepsilon}\left(\eta_K(E_H + E_2)\right) + \varepsilon\left(E_2(E_H + E_K) + E_K E_H\right) = \tilde{\sigma}(E_H + E_K)$$

$$\dot{\varepsilon} + \varepsilon \underbrace{\frac{\left(E_2(E_H + E_K) + E_K E_H\right)}{\eta_K(E_H + E_2)}}_{a} = \tilde{\sigma} \underbrace{\frac{(E_H + E_K)}{(E_H + E_2)}}_{b}$$

$$\dot{\varepsilon} + a\varepsilon = b$$

* Multiply on both sides with Integration factor

$$\dot{\varepsilon} e^{ab} + a\varepsilon e^{ab} = b e^{ab}$$

$$\frac{d}{dt}\left(\varepsilon e^{ab}\right) = b e^{ab}$$

$$\text{I.F} = e^{\int a\,dt}$$

$$\text{I.F} = e^{at}$$

* Integrating on both sides

$$\int \frac{d}{dt}(\varepsilon e^{at}) = \int b e^{at}$$

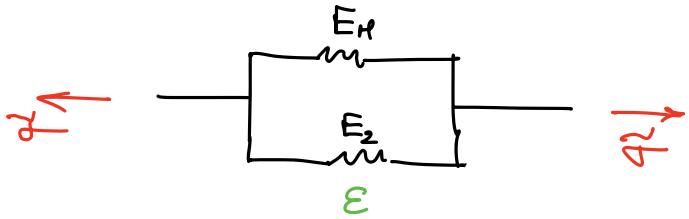
$$\varepsilon e^{at} = \frac{b}{a} e^{at} + C \longrightarrow \text{Constant of integration}$$

$$\varepsilon(t) = \frac{b}{a} + C e^{-at}$$

INITIAL CONDITION:

$$\varepsilon(0) = ?$$

* We know $\varepsilon_K(0) = 0 \Rightarrow$ physically we can imagine this as the damper providing infinite resistance when loaded suddenly, therefore the entire block behaves like a rigid body, and the network reduces to



$$\varepsilon(0) = \frac{\tilde{\sigma}}{E_H + E_2}$$

* Finally, we obtain

$$\varepsilon(0) = \frac{b}{a} + C(1)$$

$\therefore$

$$C = \varepsilon(0) - \frac{b}{a}$$

$$C = \frac{\tilde{\sigma}}{E_H + E_2} - \frac{\tilde{\sigma}(E_H + E_K)}{\eta_K\left(E_2(E_H + E_K) + E_K E_H\right)}$$

* But it may not always be possible to integrate analytically. In these cases, numerical integration is required.

• Approximating derivatives numerically:

→ Let $f(x)$ be a function, whose derivative at a point $x_0 \longrightarrow f'(x_0)$ is required. Let h be a small displacement from x_0

* $$f(x_0+h) = f(x_0) + h\frac{f'(x_0)}{1!} + \frac{h^2 f''(x_0)}{2!} + \frac{h^3 f'''(x_0)}{3!} + \dots \longrightarrow (A)$$

$$f(x_0+h) \approx f(x_0) + h\frac{f'(x_0)}{1!} + O(h^2) \longrightarrow \text{Negligible when } h \to 0$$

$$f'(x_0) \approx \frac{f(x_0+h) - f(x_0)}{h} \longrightarrow \text{Forward difference}$$

* $$f(x_0-h) = f(x_0) - h\frac{f'(x_0)}{1!} + \frac{h^2 f''(x_0)}{2!} + \frac{h^3 f'''(x_0)}{3!} + \dots \longrightarrow (B)$$

$$f(x_0-h) \approx f(x_0) - h\frac{f'(x_0)}{1!} + O(h^2)$$

$$f'(x_0) = \frac{f(x_0) - f(x_0-h)}{h} \longrightarrow \text{Backward difference}$$

* Second order derivatives:

$$f(x_0+h) + f(x_0-h) \approx 2f(x_0) + 2\frac{h^2 f''(x_0)}{2!} + O(h^4)$$

$$f''(x_0) \approx \frac{f(x_0+h) + f(x_0-h) - 2f(x_0)}{h^2}$$

Backward Euler Time Integration:

$$\varepsilon \longrightarrow \varepsilon_{n+1}$$
$$\sigma \longrightarrow \sigma_{n+1}$$
$$\dot{\varepsilon} \longrightarrow \frac{\varepsilon_{n+1} - \varepsilon_n}{\Delta t}$$
$$\dot{\sigma} \longrightarrow \frac{\sigma_{n+1} - \sigma_n}{\Delta t}$$

* Substitute all of this into Equation (7) and solve for σ_{n+1} or ε_{n+1}
* Variables that need to be known σ_n, ε_n (from previous step)

Model 1:

$$\dot{\sigma}\eta_K + \sigma(E_H + E_K) = \varepsilon\left(E_2(E_H + E_K) + E_K E_H\right) + \dot{\varepsilon}\left(\eta_K(E_H + E_2)\right)$$

$$\left(\frac{\sigma_{n+1} - \sigma_n}{\Delta t}\right)\eta_K + \sigma_{n+1}(E_H + E_K) = \varepsilon_{n+1}\left(E_2(E_H + E_K) + E_K E_H\right) + \frac{\varepsilon_{n+1} - \varepsilon_n}{\Delta t}\left(\eta_K(E_H + E_2)\right)$$

Solving for ε_{n+1}, we obtain:

$$\varepsilon_{n+1} = \frac{\varepsilon_n\left(\eta_K(E_2 + E_H)\right) - \sigma_n\eta_K + \sigma_{n+1}\left(\eta_K + \Delta t(E_H + E_K)\right)}{\Delta t\left(E_2(E_H + E_K) + E_K E_H\right) + \eta_K(E_H + E_2)}$$

$\varepsilon_n, \sigma_n \longrightarrow$ Known from previous time step

$\Delta t \longrightarrow$ chosen by us.

$\sigma_{n+1} = \sigma(t_{n+1}) \longrightarrow$ given by our chosen loading function

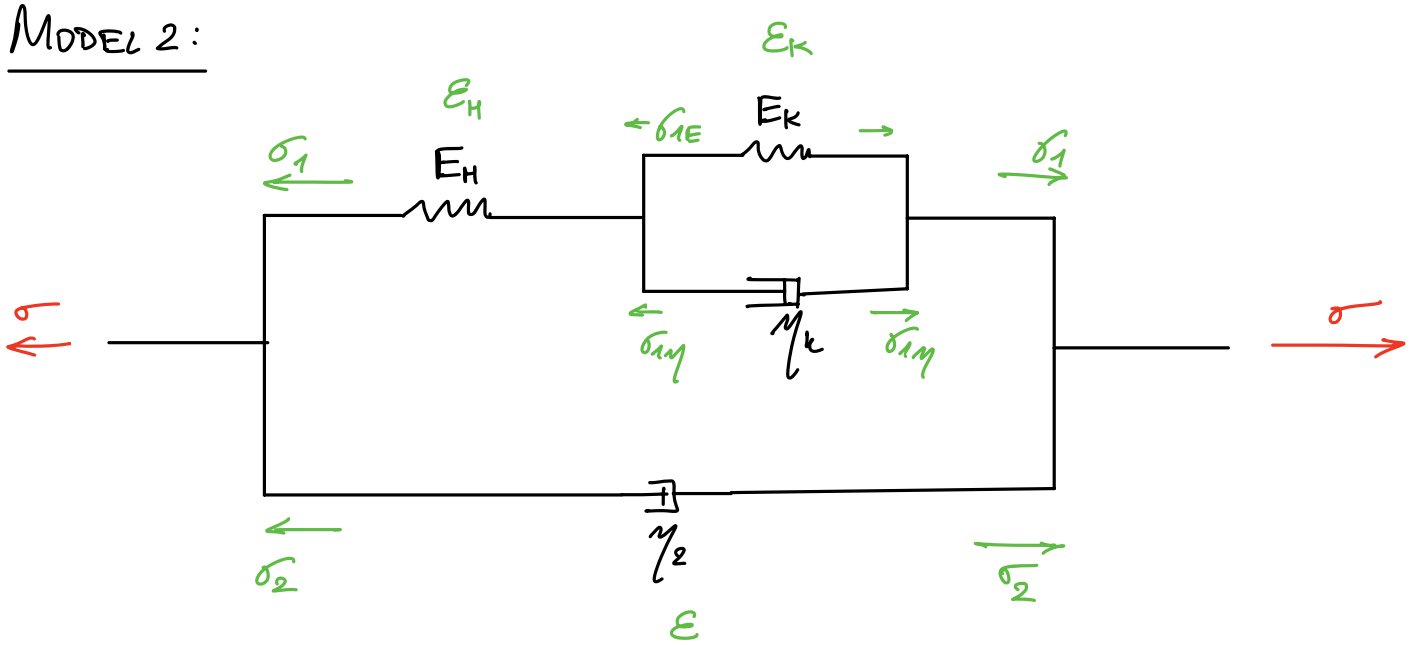
Example:

* For a load controlled problem, where we know $\sigma(t)$, we calculate

$\sigma_0 = \sigma(t_0 = 0)$, the corresponding $\varepsilon_0 = \frac{\sigma_0}{E_H + E_2}$ (See Initial conditions)

Solve for ε_{n+1}

Model 2:



* Since branch 1 is the same as in Model 1, we can use (6)

$$\dot{\sigma}_1(\eta_K) + \sigma_1(E_H + E_K) = \dot{\varepsilon}(E_H\eta_K) + \varepsilon(E_K E_H)$$

* In branch 2:

$$\sigma_2 = \eta_2\dot{\varepsilon} \qquad \sigma_1 = \sigma - \sigma_2 = \sigma - \eta_2\dot{\varepsilon}$$

$$\dot{\sigma}_2 = \eta_2\ddot{\varepsilon} \qquad \dot{\sigma}_1 = \dot{\sigma} - \dot{\sigma}_2 = \dot{\sigma} - \eta_2\ddot{\varepsilon}$$

* Inserting into (6)

$$(\dot{\sigma} - \eta_2\ddot{\varepsilon})(\eta_K) + (\sigma - \eta_2\dot{\varepsilon})(E_H + E_K) = \dot{\varepsilon}(E_H\eta_K) + \varepsilon(E_K E_H)$$

$$\dot{\sigma}\eta_K + \sigma(E_H + E_K) = \ddot{\varepsilon}(\eta_2\eta_K) + \dot{\varepsilon}(\eta_K E_H + \eta_2(E_H + E_K)) + \varepsilon(E_K E_H)$$

* Solve using backward Euler and $\ddot{\varepsilon} = \dfrac{\varepsilon_{n+1} + \varepsilon_{n-1} - 2\varepsilon_n}{(\Delta t)^2}$